

Centralized Treasury Take-Home — Question 1

What drives the S&P 500 equity financing spread over SOFR?

Albert Wong | Regression analysis of financing_spreads.csv (Jan 2020 – Dec 2025) against public macro/vol features

Bottom Line

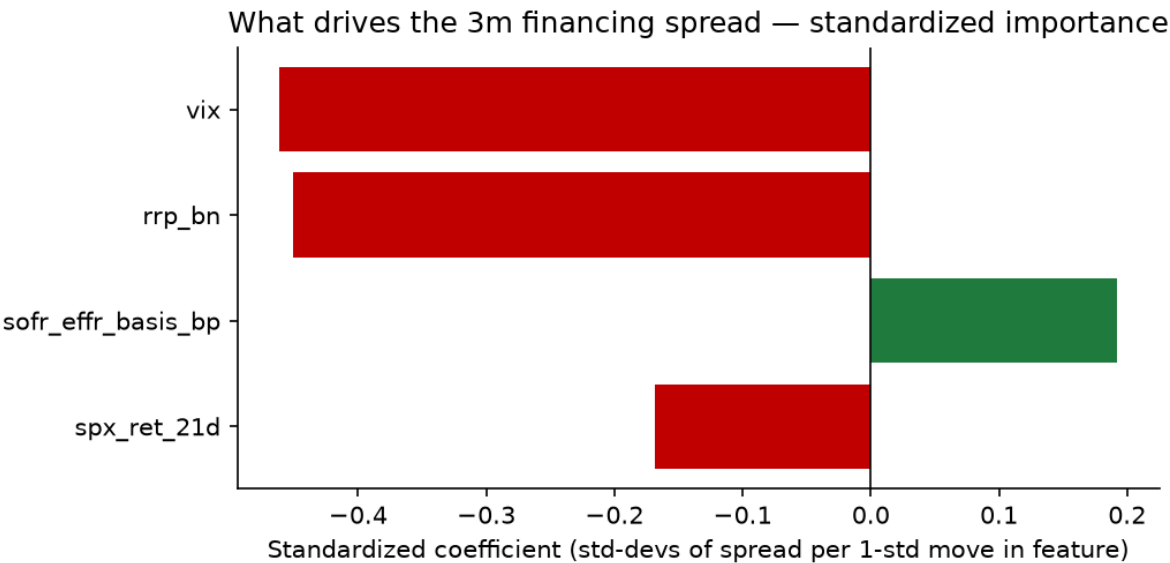
The 3-month financing spread is best explained by four features: **VIX**, the **SOFR–EFFR basis**, **ON RRP usage**, and **trailing 1-month S&P 500 return** ($R^2 = 0.46$). The spread is fundamentally **demand-driven, not supply-driven**: in calm, low-vol markets investors lever up into synthetic equity exposure, pushing the spread up; in risk-off/high-vol periods, deleveraging collapses demand for synthetic longs faster than dealer balance-sheet capacity shrinks, compressing the spread. As of the last observation in the data (2025-12-31), the model's fair value is **94.4bp** vs. an actual **57.5bp** — the spread is currently running about **37bp cheap** to fundamentals.

The Model

$\text{spread_3m_bp} = 92.4 - 1.61 \cdot \text{VIX} + 1.22 \cdot (\text{SOFR-EFFR basis, bp}) - 0.015 \cdot (\text{RRP usage, \$bn}) - 88.4 \cdot (\text{SPX 21-day return})$

OLS, Newey-West HAC standard errors (5 lags) to correct for daily autocorrelation in the spread. $n = 1,487$ trading days. All four coefficients significant at $p < 0.001$.

Feature	Coefficient	Std. coef.	p-value	Mechanism
VIX	-1.61	-0.46	<0.001	Vol → risk appetite for synthetic exposure (see below)
SOFR–EFFR basis (bp)	+1.22	+0.19	<0.001	Collateral/repo market stress
RRP usage (\$bn)	-0.015	-0.45	<0.001	System-wide cash/liquidity slack
SPX 21d return	-88.4	-0.17	<0.001	Risk-on momentum / deleveraging



Standardized coefficients let us compare features measured in wildly different units (VIX points vs. \$bn vs. bp) on one scale.

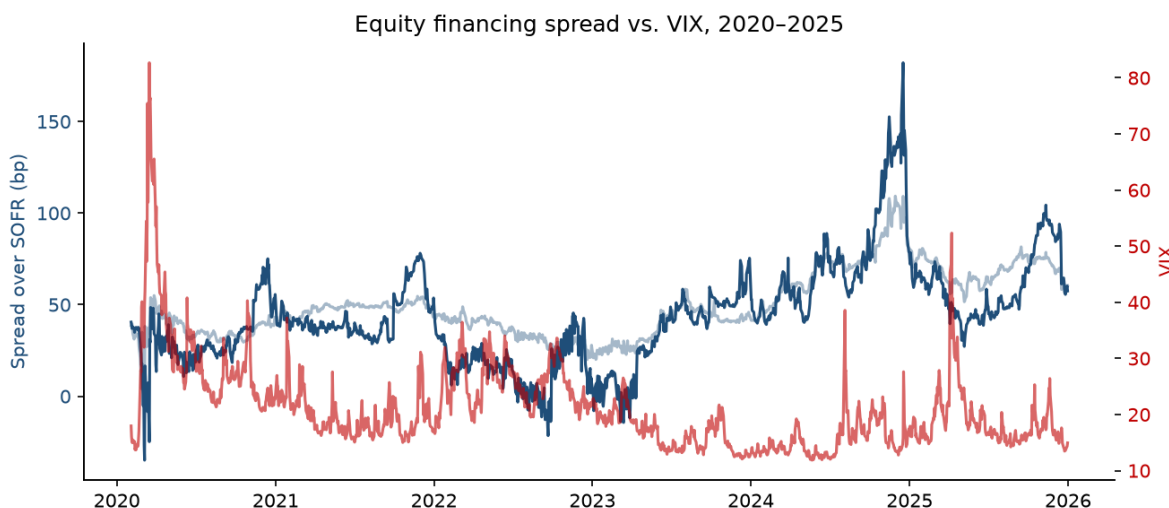
Why the 3-Month Tenor?

All four regressors are *spot/current* readings — today's VIX, today's liquidity conditions. The 3m spread reflects near-term financing conditions, so it's a clean horizon match for spot inputs. The 1y spread reflects *expectations* about financing conditions a year out — explaining that with only today's spot VIX (rather than the market's expected VIX path) is a horizon mismatch. More importantly, Question 2 is about a *tactical* decision — when to deploy capital day-to-day — which is naturally tied to the near-term rate, not a 1-year structural one. To be transparent: the 1y model is not a worse fit — it has a slightly *higher* R^2 (0.474 vs. 0.457) on the same four features. 3m is the headline tenor because it's the more decision-relevant one for what Treasury actually does with this, not because it fits the data better.

The Counterintuitive Finding: Why Is VIX Negative?

Going in, the expected sign on VIX was **positive** — higher volatility should mean higher dealer hedging/balance-sheet costs, which should be passed through as a wider spread. The data says the opposite, and it's not a fluke: the negative relationship holds (a) in the raw univariate correlation, (b) when the COVID crash window (Feb–Aug 2020) is excluded entirely, and (c) it is the *only* feature that remains significant when the model is re-estimated on day-over-day **changes** rather than levels (a robustness check run specifically because the levels regression had severe residual autocorrelation, Durbin-Watson = 0.10 — a classic warning sign that two persistent series can appear related simply because both trend over time).

The corrected interpretation: this spread is **demand-driven**. In calm markets, investors lever up and chase synthetic equity exposure because it's capital-efficient — demand for financing rises, and so does its price. In a vol spike, investors deleverage and demand for synthetic long exposure collapses faster than dealer capacity does, so the spread compresses. RRP's negative sign tells the same liquidity story: more idle cash parked at the Fed (ample system liquidity) coincides with a cheaper financing spread.

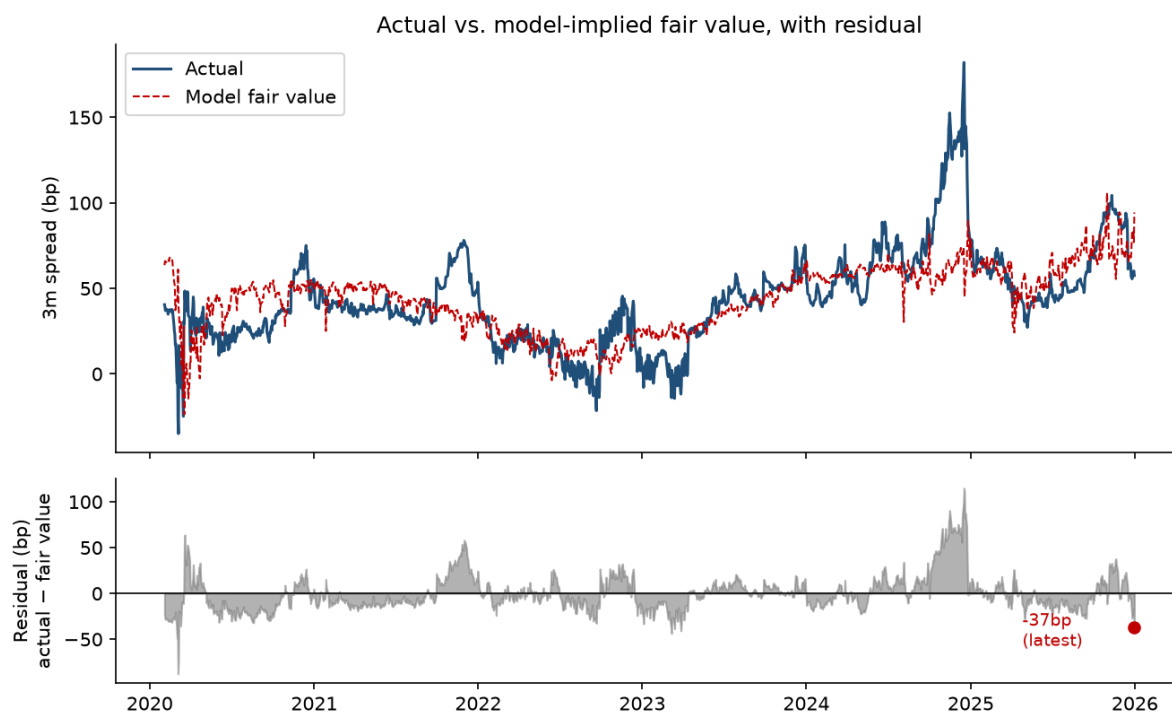


Spread (blue, 3m solid / 1y faded) vs. VIX (red). Note the inverse relationship: VIX spikes in 2020 coincide with the spread cratering (even going negative); the spread's highest levels (2024–25) occur while VIX stayed low.

Robustness Caveat

Re-estimating on day-over-day changes (rather than levels) drops R^2 to 0.04 and only VIX stays significant. The basis, RRP, and SPX-return relationships look like slow, multi-month regime co-movement (sensible — liquidity conditions and term structure evolve gradually) rather than tight daily causation. Both framings are legitimate for

different purposes, but the levels-model R^2 should be read as “what regime are we in,” not “what moved the spread today.”



Model fit over time, with the residual (actual minus fair value) shown below. The latest reading (2025-12-31, also year-end) sits 37bp below fair value — unusually cheap.

Features Considered But Not Used

Curve slope (T10Y3M) — insignificant in both the 3m ($p=0.37$) and 1y ($p=0.54$) models; dropping it cost only 0.003 of R^2 . Might matter more as a regressor on the *1y-minus-3m spread differential* itself rather than on either level.

SPX realized vol (21d) — 82% correlated with VIX (VIF 3–4) and insignificant once VIX is included; redundant, not wrong. A future cut: the *implied-minus-realized vol gap* (the vol risk premium) rather than either series alone.

Treasury General Account (TGA) level — only available weekly on FRED (WTREGEN); forward-filling to daily creates a step-function feature that understates true daily variation. A true daily TGA series exists on Treasury's Fiscal Data API (different source) and would be the cleaner fix.

Quarter-end calendar dummy — not built into this version. Notably, the dataset's last observation (2025-12-31) is itself a year/quarter-end, and its residual is unusually large (–37bp). A 0/1 flag (or days-to-quarter-end countdown) for the dealer balance-sheet “turn” effect would mainly help *clean up the residual* for a follow-on trading-signal exercise — without it, a few mechanically predictable days per quarter would otherwise look like genuine mispricing.

Data Sources Used vs. Considered

Source	Used?	Why / future path
FRED	Yes — backbone	VIX, SOFR, EFFR, RRP, SPX all sourced here. Free, reliable, no rate limits.
CFTC Commitments of Traders	No	Conceptually strong (asset-manager net futures positioning = direct demand proxy) but weekly
CBOE VIX term structure	No — used spot VIX	Spot VIX already dominates the model. Term structure (VIX9D/VIX3M) could be compared against
Yahoo Finance ETF flows	No — substituted SPX returns	Xbox does not actually publish creation/redemption flow data (that's a paid feed — ETF.com)

Current Expected Financing Cost

Plugging the latest available feature values (2025-12-31) into the model: **model fair value = 94.4bp** (3m) vs. **actual = 57.5bp** — a 37bp gap. For the 1y tenor: fair value = 79.5bp vs. actual = 60.5bp (19bp gap). On both tenors, financing is currently priced **cheap** relative to what VIX, repo conditions, system liquidity, and recent equity momentum would suggest.